

Anatomy of Mode-1 Internal Solitary Waves Derived From Seaglider Observations in the Northern South China Sea

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ABSTRACT: Observations from a Seaglider, two pressure-sensor-equipped inverted echo sounders (PIESs), and a thermistor chain (T-chain) mooring were used to determine the waveform and timing of internal solitary waves (ISWs) over the continental slope east of Dongsha Atoll. The Korteweg–de Vries (KdV) and Dubreil–Jacotin–Long (DJL) equations supplemented the data from repeated profiling by the glider at a fixed position (depth ~1017 m) during 19–24 May 2019. The glider-recorded pressure perturbations were used to compute the rarely measured vertical velocity (w) with a static glider flight model. After removing the internal tide–caused vertical velocity, the w of the eight mode-1 ISWs ranged from -0.35 to 0.36 m s^{-1} with an uncertainty of $\pm 0.005 \text{ m s}^{-1}$ due to turbulent oscillations and measurement error. The horizontal velocity profiles, wave speeds, and amplitudes of the eight ISWs were further derived from the KdV and DJL equations using the glider-observed w and potential density profiles. The mean speed of the corresponding ISW from the PIES deployed at ~2000 m depth to the T-chain moored at 500 m depth and the 19°C isotherm displacement computed from the T-chain were used to validate the waveform derived from KdV and DJL. The validation suggests that the DJL equation provides reasonably representative wave speed and amplitude for the eight ISWs compared to the KdV equation. Stand-alone glider data provide near-real-time hydrography and vertical velocities for mode-1 ISWs and are useful for characterizing the anatomy of ISWs and validating numerical simulations of these waves.

SIGNIFICANCE STATEMENT: Internal solitary waves (ISWs), which vertically displace isotherms by approximately 100 m, considerably affect nutrient pumping, turbulent mixing, acoustic propagation, underwater navigation, bedform generation, and engineering structures in the ocean. A complete understanding of their anatomy and dynamics has many applications, such as predicting the timing and position of mode-1 ISWs and evaluating their environmental impacts. To improve our understanding of these waves and validate the two major theories based on the Korteweg–de Vries (KdV) and Dubreil–Jacotin–Long (DJL) equations, the hydrography data collected from stand-alone, real-time profiling of an autonomous underwater vehicle (Seaglider) have proven to be useful in determining the waveform of these transbasin ISWs in deep water. The solutions to the DJL equation show good agreement with the properties of mode-1 ISWs obtained from the rare in situ data, whereas the solutions to the KdV equation underestimate these properties. Seaglider observations also provide in situ data to evaluate the performance of numerical simulations and forecasting of ISWs in the northern South China Sea.

KEYWORDS: Ocean; Internal waves; Solitary waves; In situ oceanic observations; Oceanic profilers

1. Introduction

Internal solitary waves (ISWs), which are evolved from the energy conversion of westward-propagating internal tides generated on the two ridges in the Luzon Strait (Fig. 1a), are easily observed in the northern South China Sea (SCS). The dynamics of the generation, propagation, and dissipation of ISWs have been intensively examined through field observations using echo sounders, thermistor chains, current profilers, and satellite remote sensing over the past two decades (Ramp et al. 2004;

Yang et al. 2004; Klymak et al. 2006; Ramp et al. 2010; Alford et al. 2010; Chang et al. 2011; Li and Farmer 2011; Farmer et al. 2011; Simmons et al. 2011; Lien et al. 2014; Alford et al. 2015; Huang et al. 2016; Chen et al. 2019). They have also been studied using theoretical analyses and numerical simulations (Vlasenko et al. 2012; Grimshaw and Helfrich 2018; Cai et al. 2002; Buijsman et al. 2010; Warn-Varnas et al. 2010; Zhang et al. 2011; Wang et al. 2020). Research interests include the physical processes involved in pumping nutrients, enhancing turbulent mixing, altering acoustic propagation, interfering with underwater navigation, causing bedform generation, and damaging engineering structures in the ocean (Osborne et al. 1978; Yang et al. 2004; Lien et al. 2005; Wang et al. 2007; St. Laurent 2008; Jan and Chen 2009; Ledwell et al. 2011; Reeder et al. 2011).

Through decades-long endeavors, the characteristics and dynamics of these dramatic large internal waves have been

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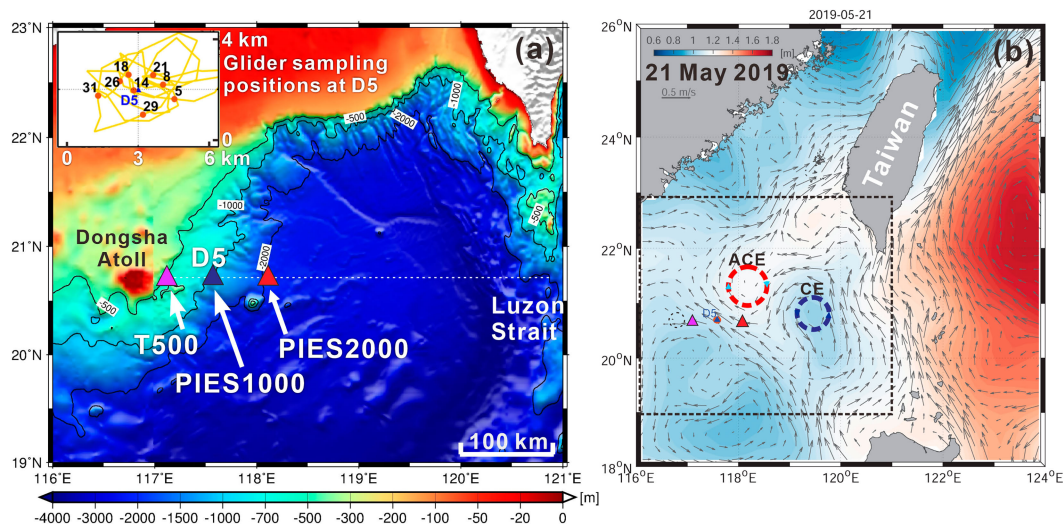


FIG. 1. (a) Bathymetric chart of the northern South China Sea. The positions as the Seaglider encountered ISWs at station D5 are shown by red dots in the upper-left inset with distances indicated by kilometers in the zonal and meridional directions. (b) AVISO SSH (color shading) and absolute geostrophic current (arrows) for 21 May 2019, which shows locations of cyclonic (CE) and anticyclonic (ACE) eddies.

mostly revealed. Yang et al. (2004), among the few early ISW studies in the northern SCS, addressed the characteristics of ISWs observed at 426 m depth northeast of Dongsha Atoll (Fig. 1) using a moored acoustic Doppler current profiler (ADCP) and thermistor chain. These ISWs were primarily mode-1 depression waves with a maximum horizontal velocity of 2.4 m s^{-1} and an isotherm vertical displacement of 106 m. They classified 41 observed ISWs into four types: one fit the Kortweg-de Vries (KdV) equation and was called type 1, two other types converted from depression to elevation waves, and the fourth consisted of mode-2 solitary waves. The type-1 ISWs were normally observed during neap tide at locations away from where depression waves would be converting to elevation waves. Their temporal variations in velocity and temperature were symmetric and the current directions were opposite between the upper and lower layers. The ISWs observed by Yang et al. (2004) actually evolved from the continental slope to shelf, and thus presented various types of properties. It is found that the nonlinearity and rotation effect modify the characteristics of ISWs from their linear theories (Gerkema et al. 2006; Grimshaw and Helfrich 2008; Helfrich and Grimshaw 2008; Alford et al. 2010). Therefore, field observations remain crucial in validating the ISW theories.

In Alford et al. (2010), 14 ISWs were observed from a synchronized mooring array zonally aligned in the northern SCS. They found that the ratios of the nonlinear wave speed and the linear nonrotating wave speed are 1.12–1.18 west of the Luzon Strait and 1.38 roughly at the junction of the deep basin and continental slope ($\sim 2000 \text{ m}$). The wave speeds were $2.5\text{--}3.0 \text{ m s}^{-1}$ from mooring location N1 (21.376°N , 118.59°E ; 2494 m depth) to MP1 (20.919°N , 117.895°E ; 1497 m depth), the vertical displacements varied between 25 and 120 m at MP1, and the widths of the quasi-sinusoidal waveforms ranged from 0.8 to 4 km at MP1. These observations help reveal the

temporal and spatial structure of ISWs in the northern SCS and are crucial for improving their modeling and theoretical analysis.

To advance the understanding of the evolution of ISWs before and after interacting with topography, a cooperative field campaign consisting of ship-based and moored instruments observations was conducted from the 2000 m deep water west of the Luzon Strait to the 2 m shallow water east of Dongsha Atoll during May–June 2019 (Chang et al. 2021; Ramp et al. 2022; Sinnett et al. 2022). The observations examined the dramatic processes that enhanced turbulence as the ISWs shoaled on the eastern flank of the Dongsha Atoll, which was attributed to wave breaking due to combined convection and Kelvin–Helmholtz instability (Chang et al. 2021). Four ISW shoaling regimes were categorized using the bottom slope and the comparison of the ISW induced zonal velocity u and its propagating speed c in the northern SCS (Ramp et al. 2022). At depths deeper than 500 m, the waves were transbasin ISWs with little change in form; from 500 to 300 m, the amplitude of the leading wave attenuated due to wave dispersion; within 300–250 m, wave breaking was the dominant process, and shallower than 250 m, most of the ISWs disintegrated due to wave breaking and turbulent mixing.

This study, as part of the aforementioned multiplatform field campaign, focuses on the measurement of the vertical velocity of ISWs in deep water and validation of solitary wave theories based on the KdV equation with higher-order nonlinearity and the Dubreil–Jacotin–Long (DJL) equation for nonlinear solitary waves of permanent form. Ramp et al. (2022) suggest that although ISWs are neither weakly nonlinear nor two-layer and could be modified by the Coriolis effect, solutions from the KdV or DJL equations are still helpful to understand the observed hydrographic features. The waveforms obtained from the solutions to the KdV and

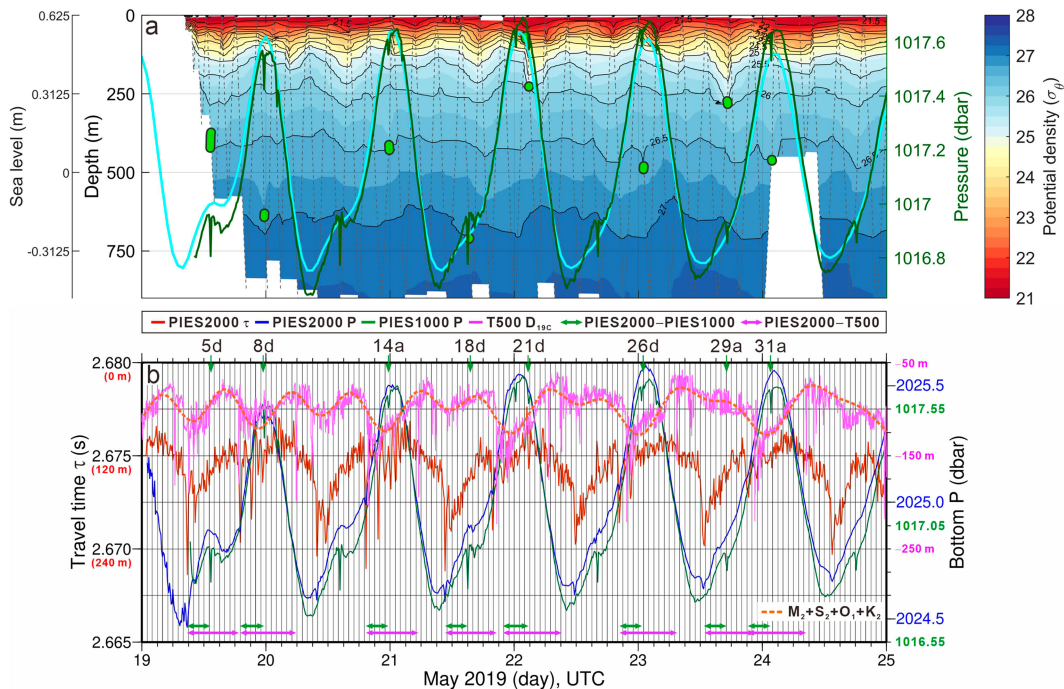


FIG. 2. (a) Potential density profiles from the Seaglider observations overlaid with bottom pressure (dark green line) at PIES1000 and tidal sea level (cyan line) from the TPXO global tide model (available at <http://volkov.oce.orst.edu/tides/>). Dotted lines are the profiling tracks. Green ovals mark the timing and depth at which the glider encountered ISWs. (b) Bottom pressure at PIES1000 (dark green line) and PIES2000 (blue line), acoustic travel time at PIES2000 (red line), and the 19°C isotherm displacement at T500 (pink line). The numbers followed by “a” (ascending) or “d” (descending) on top of (b) are the dive numbers at which the glider encountered ISWs. The corresponding thermocline displacement to the travel time change (24 m to 0.001 s) is shown on the left-hand y axis (red text). The corresponding depth of the 19°C isotherm displacement is shown on the right-hand y axis (pink text). The dashed orange line is a composite of the harmonic constants (M_2 , S_2 , O_1 , and K_1), which was obtained by applying the harmonic analysis to the 19°C isotherm displacement.

DJL equations are compared with our observations, and the performance of the two theories is evaluated. The hydrography and navigation data collected by a Seaglider deployed east of Dongsha Atoll near the 1000 m isobath are used to compute the anatomy of ISWs via the KdV and DJL equations. Seaglider observations also provide accurate arrival times and positions of ISWs for validating ISWs simulated by numerical models. The observations and methods are elaborated in the next section. Section 3 reports the characteristics and vertical velocity of each ISW observed by the glider. The solutions of the KdV and DJL equations obtained using the glider observed vertical velocity and hydrography are discussed in section 4, followed by a conclusion presented in section 5.

2. Observations and methods

The field observations were conducted along the zonal transect east of Dongsha (Fig. 1a) during May and July 2019. The meso-scale oceanic conditions during the glider observations (Fig. 1b) are shown as the satellite sea surface height (SSH) and associated absolute geostrophic current [downloaded from Archiving, Validation, and Interpretation of Satellite Oceanographic Data (AVISO) at <http://www.aviso.oceanobs.com/ducas/>]. A Seaglider

(manufactured by Kongsberg) was launched east of Dongsha (bottom depth 1017 m) on 19 May 2019 and was programmed to stay at station D5 (20.700°N, 117.583°E; Fig. 1a) for repeated profiling of the hydrography between the sea surface and 1000 m depth until 24 May 2019. The glider measured temperature, salinity, and pressure by free-flushed Sea-Bird conductivity and temperature sensors (CT Sail). The sampling rate was every 5 s (approximately every 0.75 m at an averaged glider vertical speed of 0.15 m s^{-1}) above 250 m and every 10 s (approximately every 1.5 m) below 250 m. The standard data quality control procedure (Schmid et al. 2007) was applied to the raw data. After processing, the accuracies of temperature and salinity were $10^{-3} \text{ }^\circ\text{C}$ and 0.01 psu (0.03 psu in strong thermocline regions), respectively, and the temperature and salinity resolutions were $10^{-4} \text{ }^\circ\text{C}$ and 10^{-3} psu, respectively.

Among the 31 dive/climb pairs (dives hereafter) of the observations in an area of $\sim 3 \text{ km} \times 2 \text{ km}$ (upper-left inset in Fig. 1a), the pressure records suggest that the Seaglider was forced downward and upward during 8 of the 31 dives as indicated by the green ovals in Fig. 2a. The potential density calculated from the 31 glider dives, smoothed and binned at 2 m intervals, is shown in Fig. 2a for reference. We focus on the

underlying cause for abrupt vertical displacement of the glider. Rudnick et al. (2013) showed that even a slow-moving glider can be used to observe high-frequency internal waves. They suggested that the glider can directly measure the vertical velocity of ISWs from its abrupt vertical motion during profiling. Ma et al. (2018) also observed one ISW event by an underwater glider navigated in the northern SCS. Therefore, the cause of the eight events is attributed to the vertical velocity of the passing mode-1 ISWs. The anatomy of the eight ISWs is derived from the glider observed hydrography and supplemented by the KdV and DJL equations later.

Concurrent mooring observations using two upward-looking pressure sensor-equipped inverted echo sounders (PIESs) deployed at depths of 2025 m (station PIES2000 in Fig. 1; 20.6973°N, 118.0721°E) and 1017 m (station PIES1000; 20.7045°N, 117.5924°E) were used to compute the wave speed of these ISWs. The ISW-induced large displacements in the primary thermocline were associated with perturbations in the bottom pressure and acoustic travel time, which could be observed by the PIES (Li et al. 2009). The bottom pressure was measured by a Digi-quartz pressure sensor every 10 min, and the two-way travel time was recorded by 16 pings burst-sampled every 10 min. The 10 min sampling rate could capture the mode-1 ISW signals while they were still evolving nonlinearly in deep water before interacting with the bottom (Ramp et al. 2022). The acoustic travel time at PIES2000 is used to estimate the wave speed between the two PIES stations. The acoustic travel time recorded by the PIES at PIES1000 is unfortunately too noisy to supplement the estimate.

The temperature profiles observed by a subsurface thermistor chain moored at a depth of 500 m (station T500 in Fig. 1; 20.6997°N, 117.0831°E) were additionally used to estimate wave speed from PIES2000 and PIES1000 to T500. The vertical displacement of the 19°C isotherm at T500 is selected to identify the arrival time of the ISWs because this temperature was close to the nodal point of the mode-1 ISWs there and rarely reached the surface and bottom during the observation (Ramp et al. 2022). The 1-min interval time series of the isotherm displacement is further used to fit the waveform of lower-frequency internal tides using a harmonic analysis (Foreman et al. 1993), which is in turn used to compute the vertical velocity of internal tides during the eight events. The distances from PIES2000 to PIES1000 and from PIES1000 to T500 are 50.127 and 53.265 km, respectively. The detailed configurations of the two PIESs and the thermistor chain observations and essential interpretations of the data can be found in Ramp et al. (2022) and are thus not repeated here for brevity.

The time series of the bottom pressure (dark green line) at PIES1000 and tidal sea level (cyan line) obtained from a global tide model product TPXO (<http://volkov.oce.orst.edu/tides/>) are shown in Fig. 2a. Figure 2b shows the time series of bottom pressure at PIES1000 (green line) and PIES2000 (blue line), the acoustic travel time at PIES2000 (red line), and the 19°C isotherm displacement at T500 (magenta line).

The vertical velocity of seawater (w_w) was estimated by subtracting the predicted Seaglider flight speed in static water (w_{stdy}) from the glider's vertical velocity (w_{obs}) by

$$w_w = w_{\text{obs}} - w_{\text{stdy}}, \quad (1)$$

where w_{obs} is computed as dz_g/dt (z_g is determined by the glider recorded pressure) and w_{stdy} is obtained from a steady flight model (Frajka-Williams et al. 2011). The flight model solves a balance equation between drag, lift, and buoyancy forces that a glider sustains during profiling. The glider's drag and lift forces depend on its shape, the smoothness of its outer hull, and its pitch while moving in the water. The computation of these two forces is described in Eriksen et al. (2001) and Frajka-Williams et al. (2011). The buoyancy force is a function of the glider's mass (M_{sg}) and the seawater mass displaced by the glider's volume. The computation of the buoyancy force (B_{sg}) uses seawater density (ρ_w) determined by the glider-observed temperature, salinity, and pressure and the total volume of the glider (V_{sg}) obtained from the varied volume of the glider's bladder via the equation $B_{\text{sg}} = \rho_w V_{\text{sg}} - M_{\text{sg}}g$. The detailed formulae and corresponding parameters of the flight model are found in Eriksen et al. (2001), and the procedure for the estimate of vertical velocity is found in Frajka-Williams et al. (2011). Note that the computation of B_{sg} neglects thermal inertia corrections and interstitial and tail wake effects on the glider.

The vertical velocity w_w computed from Eq. (1) comprises the components from relatively high-frequency ISWs (w_{isw}), diurnal and semidiurnal internal tides (w_{it}), relatively low-frequency mesoscale eddies (w_{ed}), random perturbations induced by turbulent eddies (w_{e}), and measurement error (m_{err}):

$$w_w \sim w_{\text{isw}} + w_{\text{it}} + w_{\text{ed}} + w_{\text{e}} + m_{\text{err}}. \quad (2)$$

The estimate of w_{isw} (approximately $w_w - w_{\text{it}} - w_{\text{ed}} - w_{\text{e}}$) is crucial for solving the velocity structure and amplitude of ISWs using the KdV and DJL equations. To estimate w_{it} , the internal tide-induced isotherm displacement at T500 is estimated using the harmonic analysis at the 19°C displacement shown in Fig. 2b. The harmonic constants of the four principal semidiurnal and diurnal constituents M_2 , S_2 , O_1 , and K_1 for the internal tides in the northern SCS (Jan et al. 2008) are used to composite the vertical displacement of the internal tides, which is illustrated as the dashed orange line in Fig. 2b. The mean w_{it} during the heave or depression of the 19°C isotherm is approximately $\pm 2.5 \times 10^{-3} \text{ m s}^{-1}$ computed using the amplitude and associated evolving time of the internal tide-induced isotherm oscillations (maximum $\sim 55 \text{ m}$), and w_{it} at the time when the glider encountered the ISWs is used to estimate w_{isw} . The estimated w_{it} is consistent with that derived from the numerical simulation of the internal tides (Chao et al. 2007; Ko et al. 2008; Ma et al. 2013). The vertical velocity of mesoscale eddies w_{ed} is typically 10^{-4} m s^{-1} (Viúdez 2018), which is three orders of magnitude smaller than the w_{isw} observed by Ma et al. (2018) and is therefore neglected in the estimate of w_{isw} . The turbulent eddy induced vertical velocity w_{e} could be 0.04 m s^{-1} , estimated from Seaglider observations in the Faroe Bank Channel at approximately 7°–8°W and 61.5°N (Beaird et al. 2012). Compared with the oscillatory duration of w_{isw} of $O(20) \text{ min}$, these very high frequency turbulent

TABLE 1. Variables of the eight ISW events derived from Seaglider's pressure data. $w_{\max}$ (m s^{-1}) is the maximum vertical velocity, $t_{\max}$ and $z_{\max}$ (m) are the time and depth corresponding to $w_{\max}$, t_{w0} is the time when $w = 0$, δt is the evolving time when the glider encountered a passing ISW, h_g is the apparent vertical displacement from the glider's pressure record, δh is net vertical displacement of the glider calculated from Eq. (3) with an uncertainty of ± 6 m, and δx ($= \delta t \times \text{linear wave speed}$) is the horizontal length scale of the ISW.

Dive No.	Lon ($^{\circ}\text{E}$)	Lat ($^{\circ}\text{N}$)	Descending $w_{\max}/z_{\max}$ ($t_{\max}$)	Ascending $w_{\max}/z_{\max}$ ($t_{\max}$)	t_{w0} (UTC)	δt	h_g (m)	δh (m)	δx (km)
5	117.5822	20.6981	-0.35/422 (1313:34 UTC 19 May)	0.34/368 (1318:47 UTC 19 May)	1315:37	27 min 5 s	53	114	2.828
8	117.5840	20.7020	-0.31/642 (2337:43 UTC 19 May)	0.22/631 (2342:33 UTC 19 May)	2340:03	23 min 50 s	11	61	2.660
14	117.5830	20.6924	-0.28/429 (2350:13 UTC 20 May)	0.36/413 (2355:00 UTC 20 May)	2352:43	19 min 28 s	16	102	2.208
18	117.5779	20.7172	-0.15/711 (1511:40 UTC 21 May)	0.15/705 (1518:12 UTC 21 May)	1514:17	25 min 4 s	6	49	2.647
21	117.5807	20.7040	-0.15/227 (0244:06 UTC 22 May)	0.27/226 (0251:04 UTC 22 May)	0247:55	40 min 41 s	1	137	4.467
26	117.5803	20.7089	-0.23/490 (0054:20 UTC 23 May)	0.20/481 (0059:50 UTC 23 May)	0057:10	23 min 39 s	9	68	2.526
29	117.5717	20.6857	-0.29/282 (1713:14 UTC 23 May)	0.31/273 (1719:19 UTC 23 May)	1715:34	27 min 12 s	9	116	3.019
31	117.5684	20.7013	-0.19/463 (0146:59 UTC 24 May)	0.26/460 (0152:59 UTC 24 May)	0149:45	25 min 4 s	3	74	2.828

perturbations can be simply removed from Eq. (2) by applying a low-pass filter to w_w without considerable influences on the estimate of w_{isw} . To evaluate the measurement error m_{err} , the w_w from Eq. (1), which was obtained from the 31 dives, is resampled every 10 s and then used to compute a spectrum of w_w . The time series is divided into 8 segments with 2048 data points in each segment. Each segment is multiplied by a Hamming window before applying the fast Fourier transform. According to the method described in Chang et al. (2011) and Beaird et al. (2012), the variance of the measurement error is obtained by integrating the power spectral density of the white noise from 3×10^{-2} to 5×10^{-2} Hz, and the result is $6.76 \times 10^{-6} \text{ m}^2 \text{ s}^{-2}$. The standard deviation of m_{err} for the glider observations is thus $2.6 \times 10^{-3} \text{ m s}^{-1}$, which is similar to that estimated by Beaird et al. (2012) ($\sim 2 \times 10^{-3} \text{ m s}^{-1}$). The uncertainty of w_{isw} mostly from w_e and m_{err} is estimated as $O(0.005) \text{ m s}^{-1}$.

3. Results

a. Anatomy of mode-1 ISWs derived from glider observations

The significant vertical perturbations in the glider's profiling correspond to PIES-observed bottom pressure perturbations (abrupt bottom pressure decreases in Fig. 2), particularly when the glider was gliding between 250 and 750 m depths. The profiling tracks (dashed lines in Fig. 2a) show that the vertical perturbations of the glider occurred during either dive or climb tracks. It is also seen that some of the ISWs recorded by the PIES were not captured by the glider during, for example, dives 12 and 24, presumably because the position of the glider was too deep for its body to significantly sense the ISW.

The glider observations were during spring tide. Computed from the glider-recorded pressure, the vertical displacement of the glider (h_g), its maximum vertical velocity ($w_{\max}$), and ISW-related vertical displacement (δh) for the eight events are listed in Table 1. As the glider encountered ISWs at t_{w0} , the associated depth ($z_{\max}$), time ($t_{\max}$), and time of zero velocity are also computed. The duration of each mode-1 ISW ranges between 19 and 27 min, except for dive 21 which was

~ 40 min. The corresponding vertical displacement h_g varies from 1 m (dive 21) to 53 m (dive 5). For a reference, Ma et al. (2018) reported that a glider, which was navigating in the northern SCS (~ 1100 m depth), met a depression ISW at ~ 400 m depth. The amplitude and period of this ISW were ~ 127 m and ~ 24 min, respectively, and the vertical velocity varied between -0.1 and 0.1 m s^{-1} . This vertical velocity is apparently smaller than those observed by our glider at similar depths (dives 5, 14, and 31 in Table 1).

b. Vertical velocity of the eight ISWs

Taking dive 5 as an example, Fig. 3 shows the time versus depth variability of w_{isw} . The dark blue lines in Fig. 3 suggest that the glider was depressed from approximately 1300 to 1317 UTC as it encountered the downwelling of mode-1 ISW, which yielded a negative w_{isw} . This was followed by uplifting from 1317 to 1330 UTC, which produced a positive w_{isw} . The maximum negative and positive w_{isw} estimated from Eq. (2) were -0.35 and 0.34 m s^{-1} for dive 5. The overall maximum downward and upward w_{isw} computed from the glider observations ranges between -0.15 and -0.35 and between 0.15 and 0.36 m s^{-1} , respectively, with an uncertainty of $\pm 0.005 \text{ m s}^{-1}$. The net upward displacement of the glider (δh) due to the passing ISW, i.e., from the lowest position at which the glider was depressed to the uppermost position at which the glider was uplifted by the ISW is further computed using

$$\delta h = \int_{t_{w0}}^{t_U} w_{\text{isw}}(t) dt, \quad (3)$$

where t_{w0} is when w_{isw} equaled zero and t_U is the end of the observation as the ISW passed the glider (Fig. 3). The δh of each ISW observed by the glider at different depths ranges from 49 to 137 m with an uncertainty of less than ± 6 m (Table 1). Note that the glider-derived w_{isw} and δh may not represent the maximum vertical velocity and displacement of each of the eight mode-1 ISWs.

Previous ship-based observations show that maximum amplitudes (i.e., vertical displacements) of ISWs in the deep

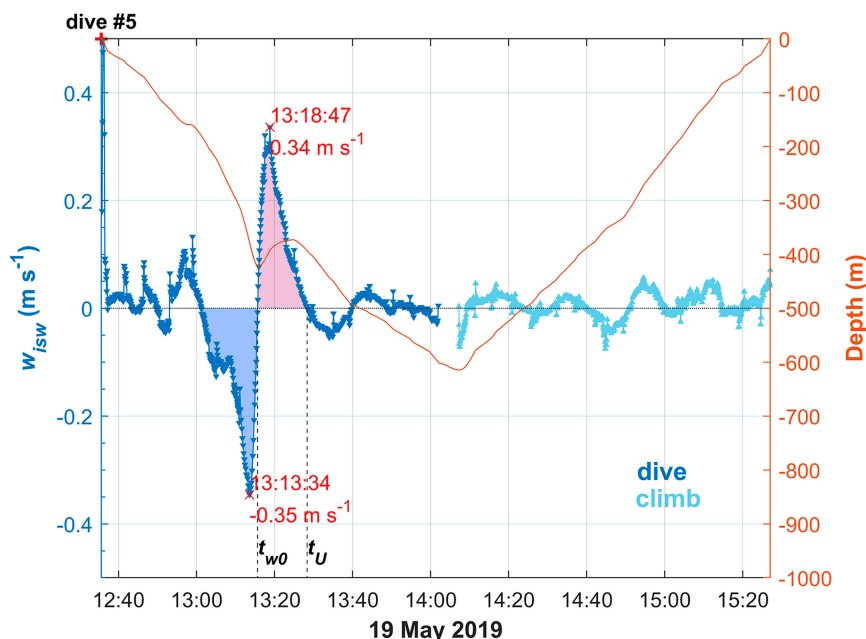


FIG. 3. Time vs depth variability of w_{isw} (dark blue line for descending and light blue line for ascending) obtained from dive 5. The uncertainty of w_{isw} is $\pm 0.005 \text{ m s}^{-1}$. The glider's depth is depicted by the red curve. Blue and pink shadings indicate negative and positive vertical velocities of the passing ISW, respectively; t_{w0} is when w_{isw} equaled zero and t_U is the time when the ISW passed over the glider.

basin ($>3000 \text{ m}$) of the northern SCS were 170 m , half wavelengths were $\sim 3 \text{ km}$, and phase speeds were $\sim 2.9 \text{ m s}^{-1}$ (Klymak et al. 2006). The maximum amplitudes of the ISWs observed by an ADCP deployed at 1497 m depth east of Dongsha were $\sim 30\text{--}110 \text{ m}$ (Alford et al. 2010). The maximum displacements and vertical velocities of the five ISWs observed

by Lien et al. (2014) at 448 and 500 m depths east of Dongsha were $106\text{--}173 \text{ m}$ and $0.23\text{--}0.71 \text{ m s}^{-1}$, respectively. Compared with these observations, the glider-observed vertical velocity and displacement of the eight ISWs were consistent with previously observed ISWs that propagated from deep water to the continental slope in the northern SCS.

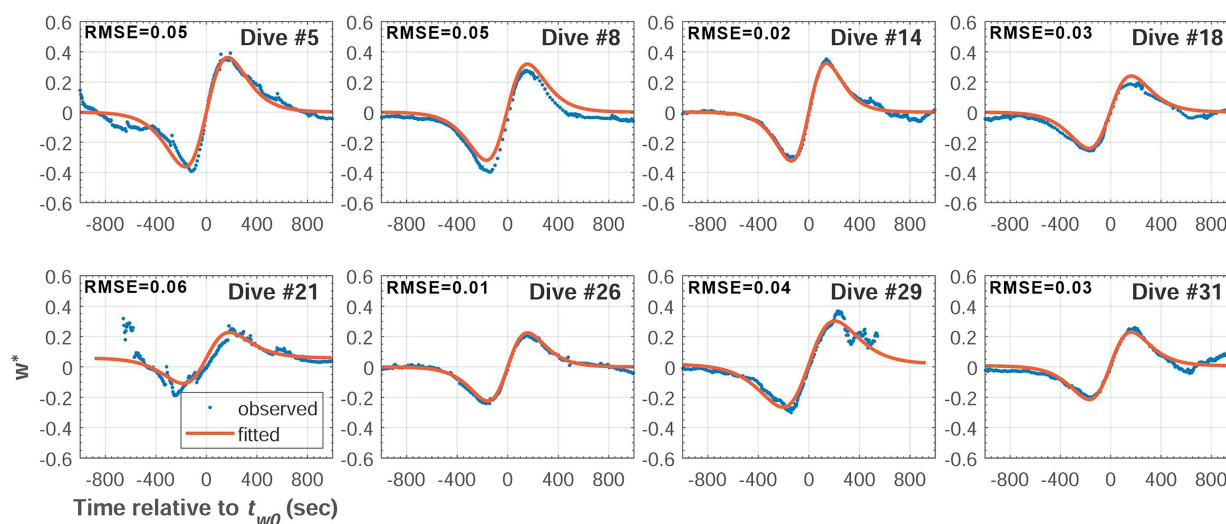


FIG. 4. Vertical velocity derived from the glider observations (blue dots) and least squares fit of normalized mode-1 vertical velocity $w^*(t)$ from Eq. (A7) (red line) for each ISW event. The root-mean-square error (RMSE) of the least squares fit is indicated in the upper-left corner of each panel.

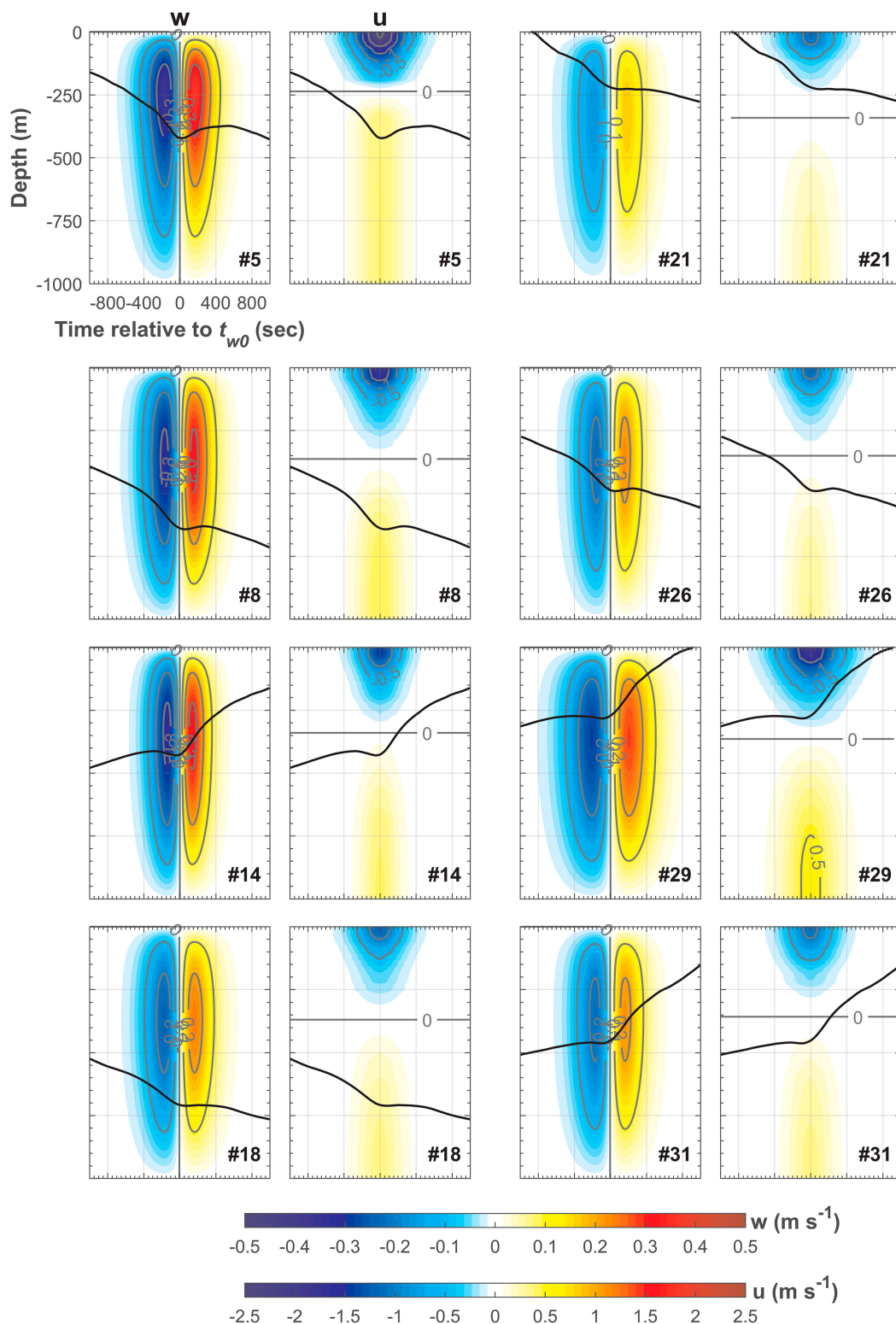


FIG. 5. Time (relative to the time of zero vertical velocity t_{w0}) vs depth varying horizontal and vertical velocities (u , w) obtained from the solutions of the KdV equation for each ISW event. The bold black line on each panel indicates the glider track. The labels for the x and y axes are only shown in the top-left panel.

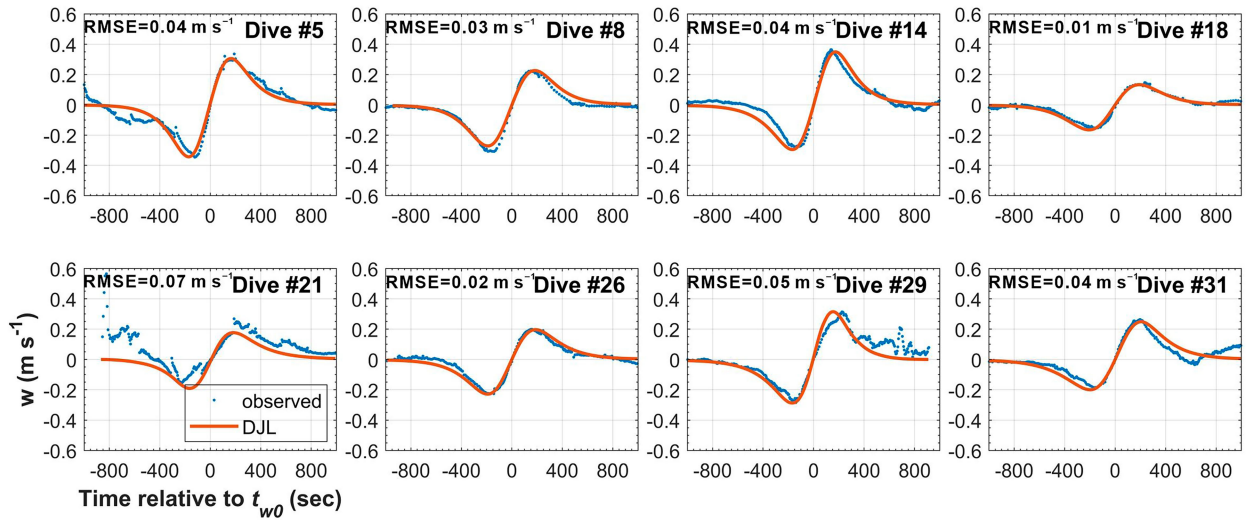


FIG. 6. As in Fig. 4, but from the solutions to the DJL equation.

4. Discussion

The vertical and horizontal velocity profiles, amplitudes, and propagation speeds of each mode-1 ISW obtained from the solutions to the KdV and DJL equations are compared with PIESs and thermistor chain observations and discussed below.

a. Solutions to the KdV equation

How the KdV equation is solved is described in appendix A. Figure 4 shows the glider-derived vertical velocity (blue dots) and least squares fit of the normalized mode-1 vertical velocity $w^*(t)$ calculated by Eq. (A7) (red curves) for each ISW. The root-mean-square errors of the fitting are 0.05, 0.05, 0.02, 0.03, 0.06, 0.01, 0.04, and 0.03 for the eight events. The horizontal and vertical velocities (u , w) derived from the solutions of the KdV equation [Eqs. (A14) and (A15)] for the eight events are shown in Fig. 5. The bold black curve on each panel in Fig. 5 represents the profiling tracks of the glider. The westward velocity is larger than 0.5 m s^{-1} in the upper 200 m and even reaches $1.5\text{--}2.0 \text{ m s}^{-1}$ for dives 5 and 29. The velocity below 200 m depth is approximately between 0.3 and 0.5 m s^{-1} eastward for each ISW. The node of the horizontal velocity profile is located approximately between 250 and 500 m for each ISW. The weakest (u , w) appears during dive 21, when the glider encountered the ISW at $\sim 230 \text{ m}$ depth. The vertical velocity of the ISW was weakened near the sea surface, and thus, the vertical movement of the glider was less influenced, which may cause larger errors in the estimate of w_{ISW} .

b. Solutions to the DJL equation

The DJL equation is described in appendix B, which includes how its solutions are obtained. Figure 6 shows the glider-observed vertical velocity (blue dots) and the least squares fit of mode-1 vertical velocity $w(t)$ obtained from the solutions to the DJL equation (red curves) for each ISW event. The root-mean-square error of each fit is 0.04, 0.03,

0.04, 0.01, 0.07, 0.02, 0.05, and 0.04 m s^{-1} for the eight sets of glider-observed w_{ISW} . Figure 7 shows the velocity (u , w) obtained from the solutions of the DJL equation. Compared with Fig. 5, the magnitudes of u obtained from DJL are normally larger than KdV. The westward velocity reaches 2.0 m s^{-1} in the upper 150 m for the ISWs observed, particularly during dive 5, 8, 14, and 18. The velocity shear derived from DJL is also stronger than that derived from KdV around the nodal depth of each horizontal velocity profile or at the trough of the depression wave. The strength of velocity shear is important because that the occurrence of shear instability could be determined by the Richardson number, computed by $R_i = N^2/u_z^2$, as $R_i < 0.25$, and the shear instability plays a crucial role on mixing, energy dissipation, and nutrient dynamics in the ocean. The estimated velocity shear at the trough of the depression wave is weaker in the solutions to the KdV equation than to the DJL equation. The larger horizontal velocities associated with the DJL solution would create more shear instabilities.

c. Mode-1 ISW speed and amplitude from observations, KdV, and DJL

The mean speed of mode-1 ISWs between the moorings at PIES2000, PIES1000, and T500 is easily estimated with the distances and arrival time differences between these stations. The three stations were aligned in a zonal section (Fig. 1a). Note that the averaged direction of the ISWs is 286° with uncertainties of $\sim \pm 27^\circ$ in this region (Ramp et al. 2010). We therefore take a mean of 286° for projecting the ISW speed estimated from the three moorings to the averaged wave direction. The projected wave speed is $\sim 97\%$ of that computed from the observations obtained by the three moorings. The error in determining the ISW propagation time is within $\pm 20 \text{ min}$ between the two PIESs and within $\pm 11 \text{ min}$ between the PIES and T500 moorings. These errors are considered in estimating the ISW speed. A mean absolute geostrophic flow of $\sim 0.2 \text{ m s}^{-1}$ oriented to the mean ISW direction (Fig. 1b) is

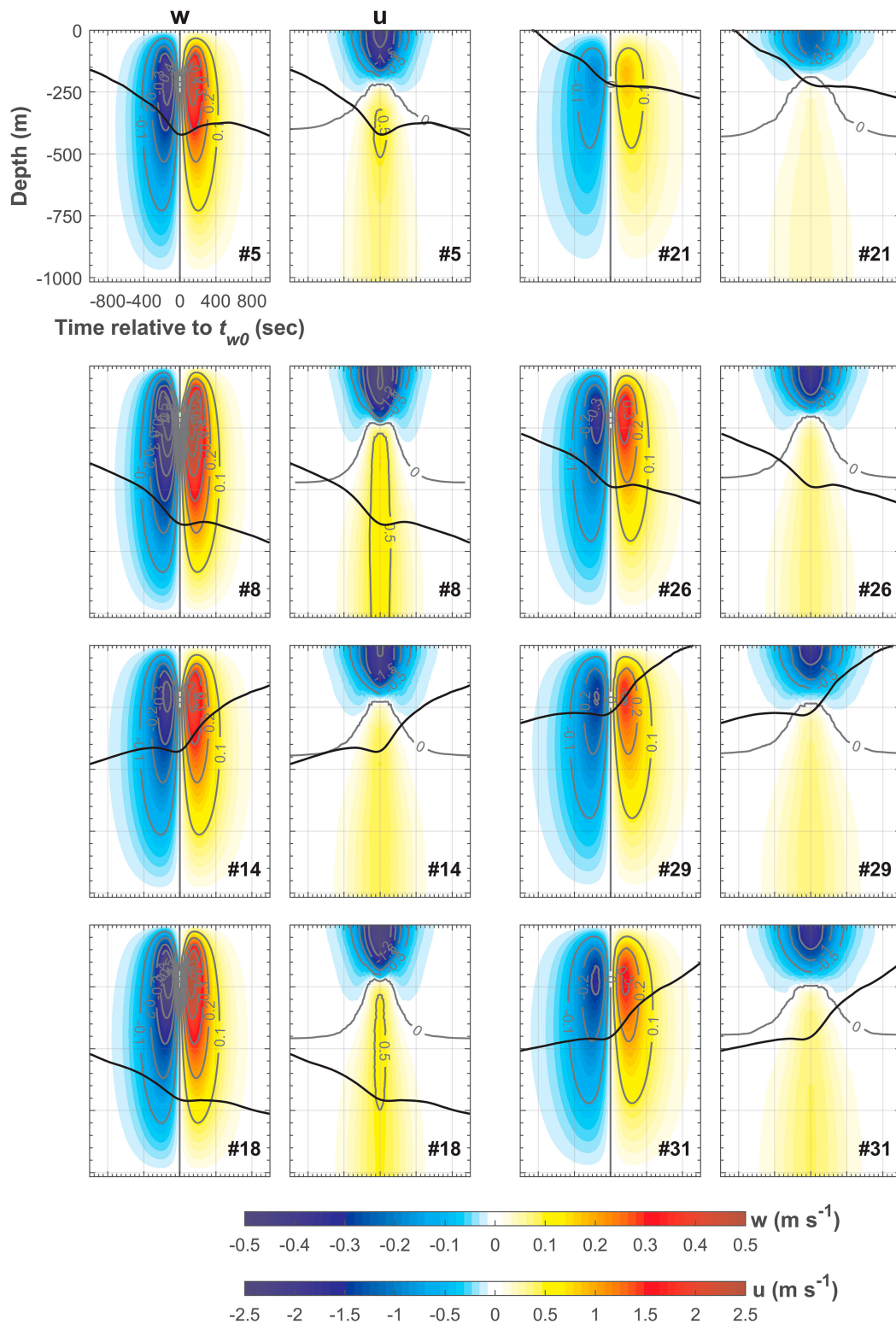


FIG. 7. As in Fig. 5, but from the solutions to the DJL equation.

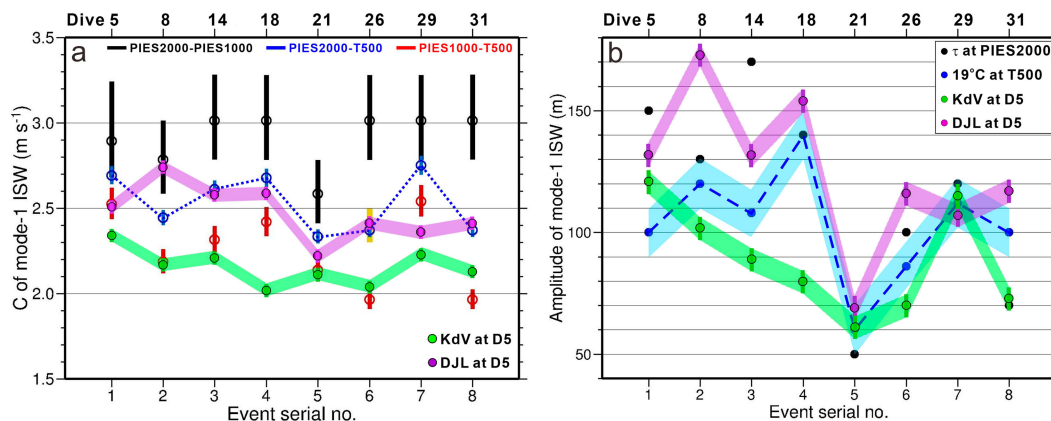


FIG. 8. (a) Wave speeds of mode-1 ISWs estimated from the mooring observations (open circles), the KdV and DJL equations (filled circles), and the MODIS image (yellow bar). The error bars for each speed computed from the observations are the measurement errors. (b) Vertical displacements (amplitudes) of mode-1 ISWs computed from the acoustic travel time at PIES2000 (black dots), the 19°C isotherm displacement at T500 (blue), and the solutions to the KdV and DJL equations. The 95% confidence intervals for wave speeds and amplitudes estimated from the solutions to the KdV and DJL equations are illustrated as green (KdV) and purple (DJL) shadings. The light blue shading represents measurement errors.

also considered in the estimate of the intrinsic speed of the ISWs. Figure 8a shows the mean speeds of the ISWs from PIES2000 to PIES1000 (black circles), from PIES2000 to T500 (blue circles), and from PIES1000 to T500 (red circles) with corresponding error bars for the eight events. It is well known that the ISW speeds decrease with decreasing bottom depth (e.g., Klymak et al. 2006; Alford et al. 2010; Ramp et al. 2019). Additionally, a 0.25-km-resolution satellite image (Fig. 9) retrieved from the Moderate Resolution Imaging Spectroradiometer (MODIS) image archive (<https://adsweb.modaps.eosdis.nasa.gov/archive/allData/61/>) is used to estimate the ISW speed during dive 26. This package of ISWs propagated at an apparent speed of

$2.4 \pm 0.1 \text{ m s}^{-1}$ (yellow bar in Fig. 8a) from D5 to where it is shown on the MODIS image. Figure 8a also shows the ISW speeds at D5 obtained from the solutions to the KdV (green) and DJL (purple) equations for comparison. The 95% confidence interval of the estimated ISW speeds (error bars in Fig. 8a) reflects the error propagation resulting from the errors in finding the solutions to the KdV and DJL equations through the least squares fit to the glider-observed w_{ISW} .

The Monte Carlo method, as described by Robert and Casella (2004), is utilized to calculate the 95% confidence interval of the wave speed and amplitude derived from the KdV and DJL equations. For each ISW event, we created 100 sets of random

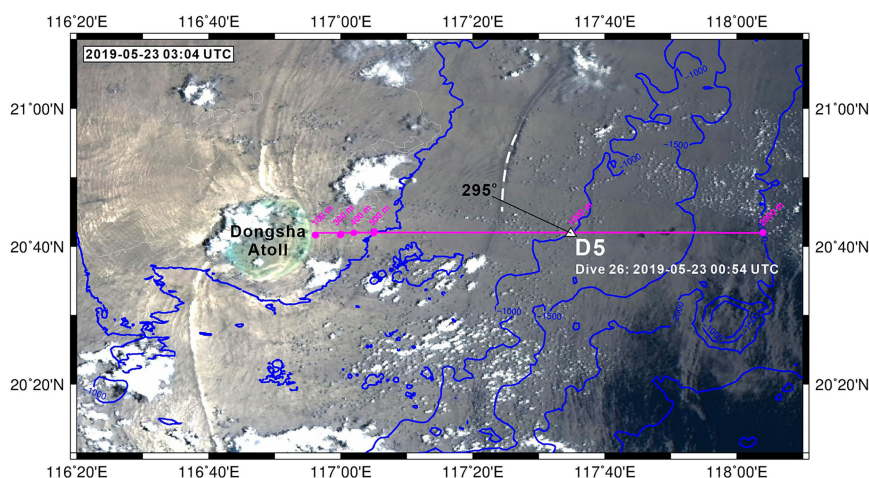


FIG. 9. MODIS image taken at 0304 UTC 23 May 2019. The white dashed curve indicates the most likely position of the package of ISWs that previously passed D5 and was encountered by dive 26 during 0054–0059 UTC on the same day. The distance from D5 to the wave crest is ~ 19.5 km. The time duration from the glider encountered the ISW at D5 to the time of this MODIS image is 130 min.

vertical velocity noise, each with a standard deviation of 0.005 m s^{-1} . This random velocity noise was added to the glider-observed w_{ISW} of each ISW. Subsequently, this glider-observed w_{ISW} with one set of random velocity noise was used to estimate the solutions to the KdV and DJL equations. This process was repeated 100 times, resulting in 100 sets of solutions for each equation and each ISW event. The 95% confidence interval of variables such as the wave speed is therefore determined by excluding the smallest and largest 2.5% (2% in practice) of the estimated wave speeds from the set of 100 solutions. The maximum range of the 95% confidence interval among the eight events is 0.08 m s^{-1} , which is used to represent the 95% confidence interval of the estimated wave speed for the eight ISWs. The same method is applied to estimate the uncertainty of the amplitudes derived from the KdV and DJL equations, and the largest range of 95% confidence interval is 10 m for the eight ISWs.

Compared with the mean ISW speed computed from PIES2000 to T500 observations (blue circle in Fig. 8a) and the one from the MODIS image (yellow dot in Fig. 8a), the eight wave speeds derived from KdV (green dots in Fig. 8a) are 6%–24% smaller than the observations. The eight wave speeds derived from DJL (purple dots in Fig. 8a) are consistent with the observations for dives 14, 18, 26, and 31, but underestimated by $\sim 7\%$, 5% , and 15% for dives 5, 21, and 29, respectively, and overestimated by $\sim 11\%$ for dive 8. The underestimation of wave speed by the KdV equation is consistent with the conclusion made by Alford et al. (2010), which is attributed to the underestimates of the KdV equation in both the increase above the linear wave speed and the discrepancy between the large and small amplitude ISWs. The better consistency of the DJL-derived wave speed with the observations, compared to the KdV-derived wave speed, could be attributed to the high nonlinearity of the DJL equation. Its full nonlinear-dispersive solutions are known to describe large amplitude mode-1 ISWs in deep water better than the KdV equation (Stastna 2022).

The wave amplitudes of the eight ISWs derived from the solutions to the KdV and DJL equations are compared with those computed from the 19°C isotherm displacement at T500 (Fig. 2b). The amplitudes obtained from KdV and DJL shown in Fig. 8b correspond to the maximum displacement of mode-1 ISWs over the 1000 m depth. Considering the 95% confidence interval in the computation, the KdV-derived amplitudes (green dots in Fig. 8b) are underestimated by $\sim 20\%$ – 43% for dives 8, 14, 18, and 31, and is overestimated by $\sim 20\%$ for only dive 5. Among the eight DJL-derived amplitudes (purple dots in Fig. 8b), five of them are approximately 18%–44% larger than the 19°C isotherm displacement at T500, while the other three are comparable in magnitude. It is noteworthy that the variability of the DJL-derived amplitudes from the first to eighth ISWs follows the variability observed in the displacement of the 19°C isotherm. The variability of the KdV-derived amplitudes, however, does not exhibit consistency with the displacement of the 19°C isotherm. Accordingly, a scatterplot in Fig. 10 shows that there is a linear relationship between the 19°C isotherm displacement and DJL-derived amplitudes for the eight ISWs at 1000 and 500 m depths. The slope of the

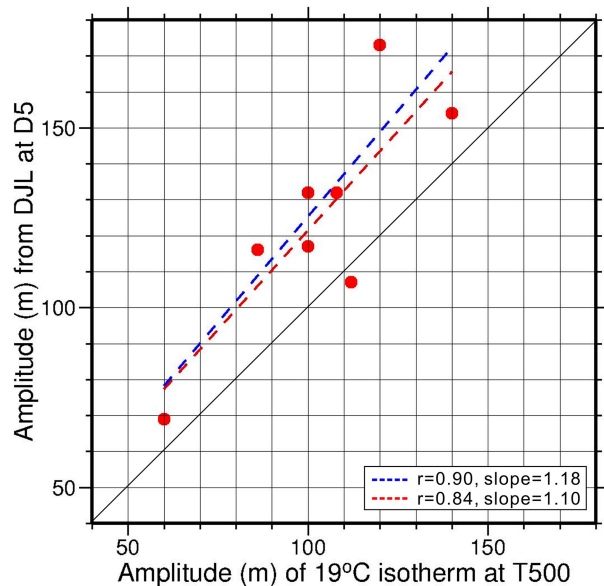


FIG. 10. Amplitudes of the 19°C isotherm displacement vs those obtained from DJL at D5. The red dashed line is the linear relationship of the eight events; the blue dashed line is the linear relationship with the point 19°C larger than DJL excluded. The r represents the linear correlation coefficient.

linear relationship is 1.10 (red dashed line in Fig. 10) for the eight events and is 1.18 (blue dashed line in Fig. 10) if the event with 19°C isotherm displacement $>$ DJL-derived amplitude is excluded.

The amplitudes estimated from the two-way travel time at PIES2000 are included in Fig. 8b for reference purposes. Note that due to the uncertainty in acoustic travel time (Li et al. 2009) and the sampling interval of 10 min, there is a potential for significant errors of greater than $\pm 30 \text{ m}$ in the corrected displacements, which makes the comparison meaningless.

Furthermore, the ISW-induced bottom pressure perturbation (p') derived from the solutions of the DJL equation is compared with that from the PIES at PIES1000, as illustrated in Fig. 11. The raw p' from the PIES observation was high-pass filtered with a cutoff period of 3 h. The timing of the maximum p' drop is shifted to 0 s on the time axis for both PIES-observed and DJL-derived p' . By comparison, the timing and waveform of p' from the PIES observations and the DJL equation for each event are consistent. The best estimate between DJL and the observations comes from dive 21. Despite the underestimation of p' from DJL for dive 5, dives 8, 14, 26, and 29 are overestimated by $\sim 50\%$ – 150% of the observed bottom p' . The inconsistency between the observed and DJL-derived p' decrease is presumably due to the low sampling time resolution (every 10 min) of the PIES observation and the inability of the pressure sensor (accuracy of 0.01% or $\sim 0.1 \text{ dbar}$ at 1000 m depth) to adequately measure the bottom pressure perturbation caused by ISWs accurately. Nevertheless, the combination of stand-alone glider observations with predictions from either the KdV or DJL equations

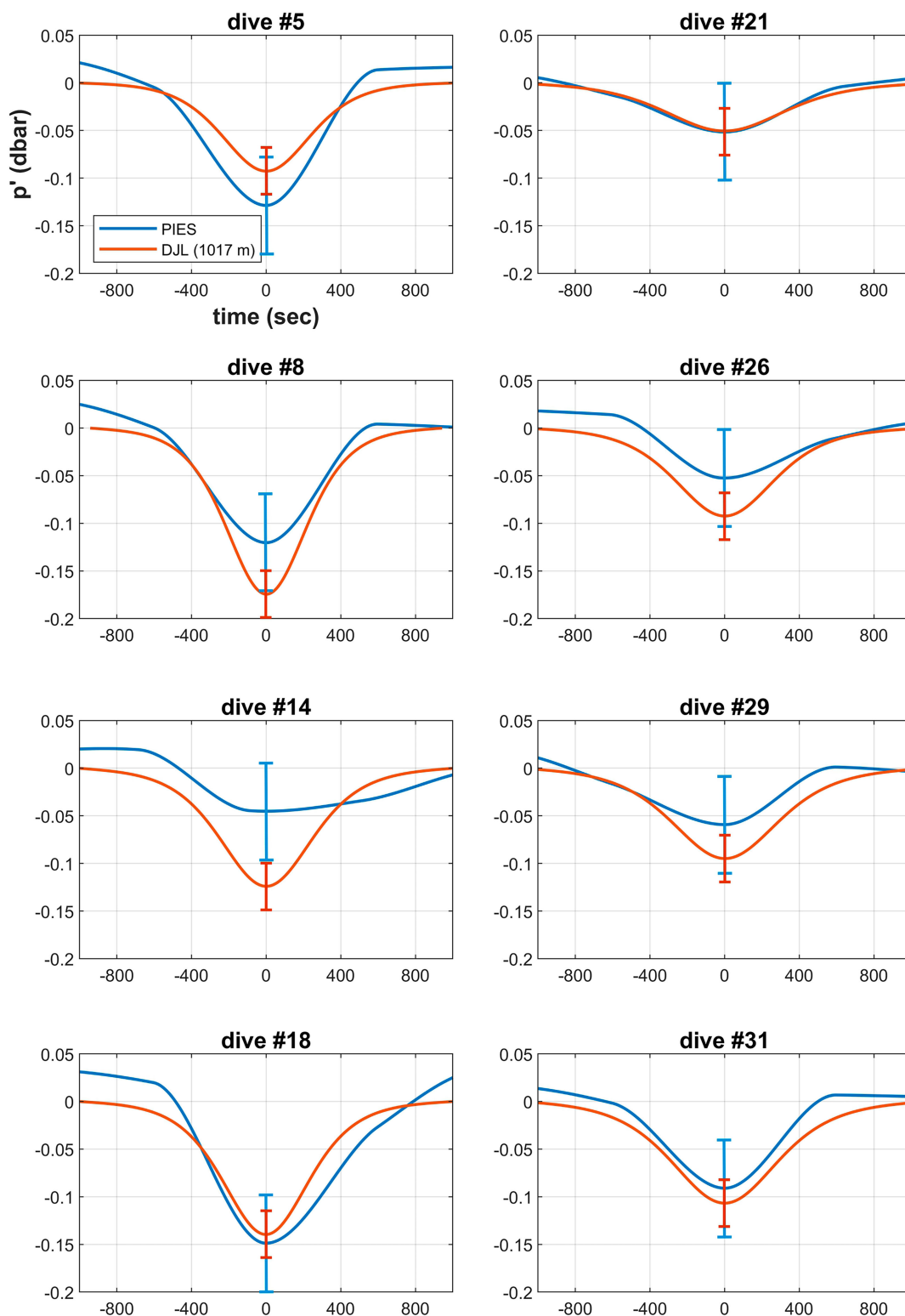


FIG. 11. PIES-observed (blue line) and DJL-derived (red line) bottom pressure perturbation (p') of ISWs for the eight events. The timing for the maximum drop of p' is set to zero for both PIES-observed and DJL-derived p' . The blue and red bars indicate the uncertainty of the PIES-observed and DJL-derived pressures, respectively. The labels for the x and y axes are only shown in the top-left panel.

has shown promise as a potential approach to fitting the waveform of transbasin ISWs in the northern SCS.

Previous efforts to predict the structure and arrival times of ISWs in the SCS include 1) addressing the relationship between the backscatter of ISW-modulated surface waves obtained by the ship or satellite radar, and its interior properties (Chang et al. 2008; Plant et al. 2010; Lund et al. 2013); 2) numerical simulations using three-dimensional, high resolution, and weakly nonlinear models (Jan et al. 2008; Simmons et al. 2011; Zhang et al. 2011; Ma et al. 2013; Alford et al. 2015); and 3) empirical model parameterizations using ISW signatures recorded in satellite imagery (Jackson 2009). However, these efforts have unavoidable restrictions. The applications of numbers 2 and 3, without appropriate in situ hydrography profiles, are only suitable to predict the propagation speed and arrival time of ISWs. The local sea surface winds and other physical processes could modulate or contaminate the radar backscatters from surface waves modulated by ISWs and then alter number 1. Uncertainty also arises due to the unpredictable presence of mesoscale eddies, changes to the Kuroshio path in the Luzon Strait, and submesoscale processes, which are poorly resolved in the numerical and empirical models. All of the above deficiencies point to a need to incorporate real-time observations into the prediction system. Our attempt to combine glider observations with associated ISW theories provides another feasible way to determine the waveform and timing of ISWs. The application of our results is encouraged.

5. Concluding remarks

Observations from a Seaglider, two PIESs, and one thermistor chain mooring during 19–24 May provide the arrival times, locations and vertical velocity of eight glider observed mode-1 ISWs in the northern SCS, which help validate the ISW theories that are based on the KdV and DJL equations. The vertical velocity and displacement of each ISW derived from the glider-observed pressure perturbations were from -0.35 to 0.36 m s^{-1} with $\pm 0.005 \text{ m s}^{-1}$ uncertainty and from 49 to 137 m with $\pm 6 \text{ m}$ uncertainty, respectively, at the depth where the glider encountered the ISW. The observations were further used to solve the wave velocity (u, w), amplitude, and phase speed of each ISW via the KdV and DJL equations. The maximum amplitude of each ISW computed from the KdV and DJL equations is between 60.6 and 172.9 m. The nonlinear phase speeds, $2.02\text{--}2.34 \text{ m s}^{-1}$ from the KdV equation and $2.22\text{--}2.74 \text{ m s}^{-1}$ from the DJL equation, increase with increasing of ISW amplitude. The wave speeds derived from the KdV equation for the eight ISWs are underestimated by 6%–24% compared to those derived from the PIES2000 and T500 observations. In contrast, the wave speeds derived from the DJL equation for the eight ISWs are closer to those derived from the observations compared to those KdV-derived wave speeds. The amplitudes of the eight ISWs obtained from DJL are also in better agreement with the corresponding 19°C isotherm displacement at T500 compared with those obtained from KdV. The velocity shears at the trough of the depression mode-1 ISW are higher in the

solutions to the DJL than to the KdV equations, leading to more shear instabilities within the core of mode-1 ISWs. The full nonlinear-dispersive solutions of the highly nonlinear DJL equation agree well with mode-1 ISWs in deep water, whereas the underestimates produced by the KdV equation in the linear wave speed and the difference between the large- and small-amplitude ISWs underestimate the ISW speed, which have been discussed in Alford et al. (2010).

The waveform derived from stand-alone glider observations in deep water, combined with an ISW theory, could be crucial for studying the evolution of ISWs as they propagate onto the shoaling topography east of Dongsha and for numerical forecasting of these waves in the northern SCS. In particular, glider observations provide real-time to near-real-time hydrographic profiles, a capability not easily achieved by moored instruments lacking real-time data transmission functionality. The waveform and timing of ISWs obtained from real-time profiling observations can be valuable in aiding the navigation of underwater vehicles.

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Data availability statement. Satellite sea surface height and associated absolute geostrophic current were downloaded from the Archiving, Validation and Interpretation of Satellite Oceanographic (AVISO) data at <http://www.aviso.oceanobs.com/ducas/>. The Moderate Resolution Imaging Spectroradiometer (MODIS) satellite image was obtained at <https://ladsweb.modaps.eosdis.nasa.gov/archive/allData/61/>. The TPXO global tide model-produced tidal sea level was downloaded from <http://volkov.oce.orst.edu/tides/>. The observational data used in this study can be downloaded freely under Mendeley at <https://doi.org/10.17632/x539yyn5n3.1>.

APPENDIX A

KdV Equation and the Analytical Solutions

The KdV equation is commonly applied to represent non-rotating, weakly nonlinear, finite-amplitude, plane-progressive ISWs propagating in a specific direction (Apel et al. 1997):

$$\frac{\partial \eta}{\partial t} + c_0 \frac{\partial \eta}{\partial x} + \alpha \eta \frac{\partial \eta}{\partial x} + \beta \frac{\partial^3 \eta}{\partial x^3} = 0, \quad (\text{A1})$$

where η is amplitude of the ISW, and c_0 is the linear wave speed. The coefficients α and β in Eq. (A1) are the quadratic

and dispersive coefficients of the ISW, respectively, which are defined as

$$\alpha = \frac{3c_0}{2} \frac{\int_{-H}^0 U^3(z) dz}{\int_{-H}^0 U^2(z) dz}, \quad \text{and} \quad (\text{A2})$$

$$\beta = \frac{c_0}{2} \frac{\int_{-H}^0 W^2(z) dz}{\int_{-H}^0 U^2(z) dz}, \quad (\text{A3})$$

where H is water depth and $U(z)$ and $W(z)$ are vertical structure functions of horizontal and vertical motion, respectively, which satisfy $U(z) = dW(z)/dz$. The analytical solutions of horizontal and vertical velocity for KdV equation are

$$u(x, z, t) = C\eta U(z) \text{sech}^2\left(\frac{x - Ct}{\Delta}\right), \quad (\text{A4})$$

$$w(x, z, t) = \frac{2C\eta}{\Delta} W(z) \text{sech}^2\left(\frac{x - Ct}{\Delta}\right) \tanh\left(\frac{x - Ct}{\Delta}\right), \quad (\text{A5})$$

where Δ represent length scale of the ISW and C is the nonlinear wave speed of the ISW defined as

$$C = c_0 + \frac{1}{3}\alpha\eta. \quad (\text{A6})$$

At a fixed position x_0 , rearranging Eq. (A5) yields the time series of vertical velocity as

$$w^*(x_0, t) = \frac{w(x_0, z, t)}{W(z)} = \frac{2C\eta}{\Delta} \text{sech}^2\left(\frac{x_0 - Ct}{\Delta}\right) \tanh\left(\frac{x_0 - Ct}{\Delta}\right). \quad (\text{A7})$$

The glider-observed hydrographic profiles of each event are used to calculate vertical velocity structure $W(z)$ and linear wave speed c_0 for mode-1 ISW. The time series of $w^*(x_0, t)$ is obtained from $w_{\text{ISW}}(z, t)/W(z)$ and Eq. (A7) is therefore alternatively written as

$$w^*(t) = A_1 \text{sech}^2(A_2 t + A_3) \tanh(A_2 t + A_3) + A_4, \quad (\text{A8})$$

$$A_1 = \frac{2C\eta}{\Delta}, \quad (\text{A9})$$

$$A_2 = -\frac{C}{\Delta}, \quad (\text{A10})$$

and A_3 and A_4 are the residuals. Rearranging Eqs. (A6), (A9), and (A10) obtains

$$\eta = -\frac{A_1}{2A_2}, \quad (\text{A11})$$

$$\Delta = -\frac{c_0}{A_2} + \frac{A_1\alpha}{6A_2^2}, \quad \text{and} \quad (\text{A12})$$

$$C = c_0 - \frac{A_1\alpha}{6A_2}. \quad (\text{A13})$$

Consequently, the velocity solutions for mode-1 ISWs become

$$u(z, t) = \left(\frac{A_1^2\alpha - 6A_1A_2c_0}{12A_2^2} \right) U(z) \text{sech}^2(A_2 t + A_3) \quad \text{and} \quad (\text{A14})$$

$$w(z, t) = A_1 W(z) \text{sech}^2(A_2 t + A_3) \tanh(A_2 t + A_3). \quad (\text{A15})$$

The coefficients A_1 , A_2 , A_3 , and A_4 are estimated using Levenberg–Marquardt algorithm for nonlinear least squares fitting (Seber and Wild 2003) of the glider observed vertical velocity (w_{ISW}/W), and their values for the eight mode-1 ISWs are listed in Table A1. The corresponding nonlinear wave speed, length scale, and amplitude of each ISW are also summarized in Table A1. To solve the KdV equation, potential density profiles and buoyancy squares (N^2) observed prior to the eight events are shown in Figs. A1a and A1b, respectively. The square of the buoyancy frequency is computed using $N^2 = -(g/\rho)[\partial\sigma_\theta(z)/\partial z]$, where g (9.81 m s^{-2}) is the gravitational acceleration. Figs. A1c and A1d present normalized mode-1 vertical and horizontal velocity profiles, respectively, estimated by normal mode theory (Pedlosky 2003).

TABLE A1. Results of Levenberg–Marquardt nonlinear least squares fit of the Seaglider observed vertical velocity ($w^* = w_{\text{ISW}}/W$), including parameters A_1 , A_2 , A_3 , and A_4 in Eq. (A14), linear wave speed c_0 , amplitude η , length scale Δ , and nonlinear wave speed C of mode-1 ISW.

Dive No.	A_1	A_2	A_3	A_4	c_0 (m s^{-1})	η (m)	Δ (m)	C (m s^{-1})
5	0.94	3.9×10^{-3}	2.8×10^{-14}	2.2×10^{-14}	−1.74	−120.7	600.1	−2.34
8	0.83	4.1×10^{-3}	0.026	2.2×10^{-14}	−1.79	−102.1	536.0	−2.17
14	0.84	4.7×10^{-3}	3.1×10^{-12}	2.2×10^{-14}	−1.89	−88.8	466.9	−2.21
18	0.62	3.9×10^{-3}	0.022	2.2×10^{-14}	−1.76	−79.7	515.2	−2.02
21	0.44	3.6×10^{-3}	2.3×10^{-14}	0.059	−1.83	−60.6	587.2	−2.11
26	0.58	4.2×10^{-3}	2×10^{-9}	2.2×10^{-14}	−1.78	−69.7	487.5	−2.04
29	0.74	3.2×10^{-3}	2.2×10^{-14}	0.018	−1.85	−114.9	693.8	−2.23
31	0.58	4×10^{-3}	0.016	7.2×10^{-3}	−1.88	−72.9	537.6	−2.13

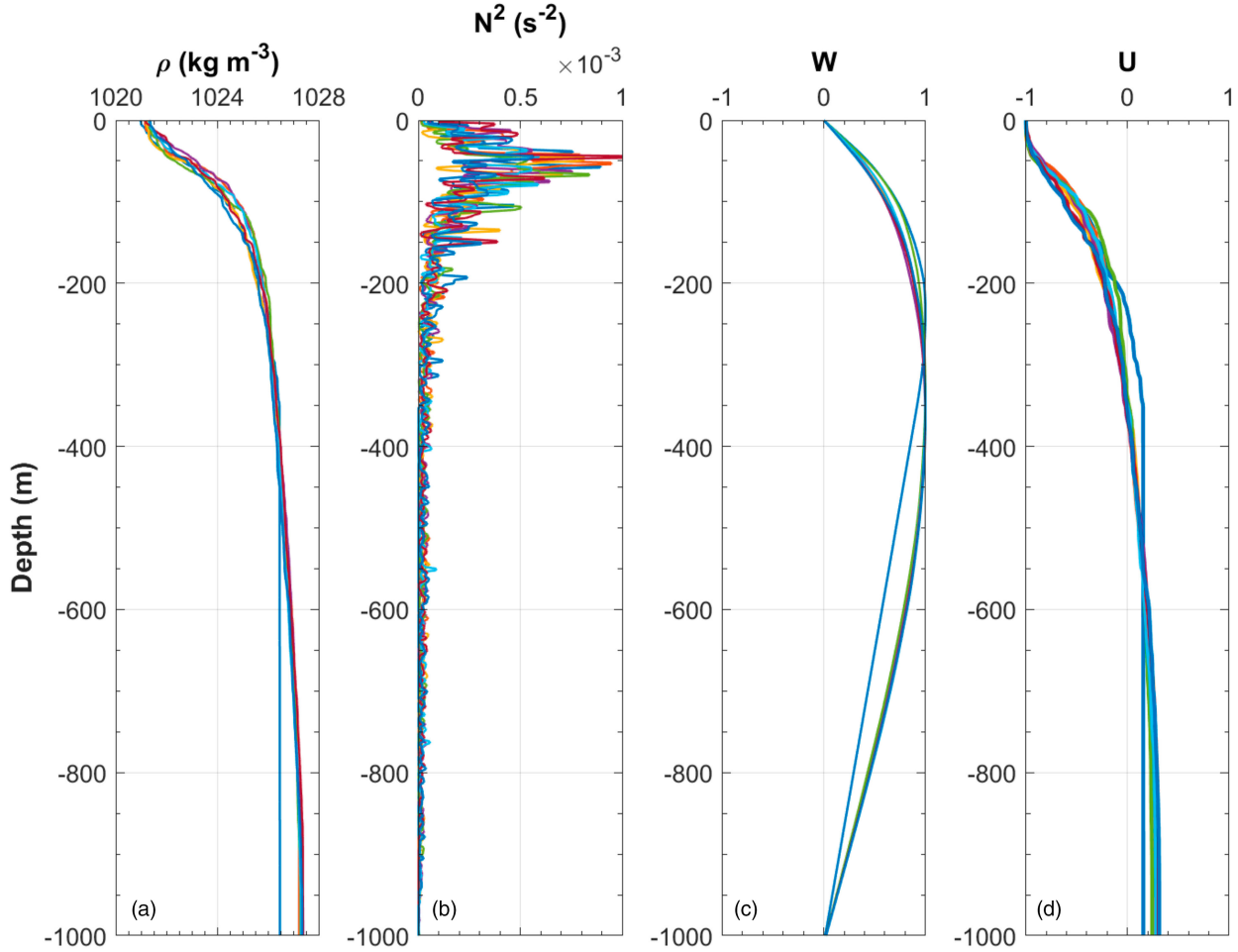


FIG. A1. Hydrography profiles observed by the Seaglider. (a) Depth-reordered potential density prior to each of the eight ISWs, (b) associated buoyancy frequency square, (c) vertical velocity structure derived from the nondimensional vertical velocity structure of the normal mode, and (d) nondimensional horizontal velocity structure of the normal mode.

APPENDIX B

DJL Equation and the Method for Finding the Solutions

The DJL equation is similar to the full set of stratified Euler equations (Stastna and Lamb 2002; Lamb 2003; Stastna 2022), which is written as

$$\nabla^2 \eta + \frac{N^2(z - \eta)}{[c_0 - U_b(z - \eta)]^2} \eta + \frac{U_{bz}(z - \eta)}{c_0 - U_b(z - \eta)} \times [\eta_x^2 + (1 - \eta_z)^2 - 1] = 0, \quad (\text{B1})$$

where $\eta(x, z)$ is amplitude of nonlinear internal waves, c_0 is wave speed, U_b is the background velocity profile, and the operator ∇^2 is defined as $(\partial^2/\partial x^2) + (\partial^2/\partial z^2)$. The equation is formally elliptic, which has no time variation and a form dominated by a Laplacian and is strongly nonlinear because that the background density and background current (if there is) are evaluated at the upstream height, i.e., at $z - \eta$ (Stastna 2022). The DJL equation can be significantly simplified if there is no background current:

$$\nabla^2 \eta + \frac{N^2(z - \eta)}{c_0^2} \eta = 0. \quad (\text{B2})$$

According to the description in Stastna (2022), Eq. (B2) can be iteratively solved within the domain of horizontal distance L and depth H . The boundary conditions are

$\eta \rightarrow 0$ in the far upstream and downstream and $\eta = 0$ at $z = 0$ and $-H$.

The associated initial guess is derived from the KdV equation and the scaled available potential energy A_{PE} is set as a constant, while the kinetic energy is minimized. The detailed iteration procedure is referred to in Stastna (2022, chapter 4). Briefly, assuming that amplitude and wave speed obtained from current iteration (n) are η^n and c_0^n , respectively, their next iteration ($n + 1$) are solved from the Poisson problem:

$$\nabla^2 \eta^n = -\lambda^n S(z, \eta^n), \quad (\text{B3})$$

where $\lambda^n = gh/c_{0,n}^2$ and $S(z, \eta) = -\bar{p}'(z - \eta)/H$. The next iteration is computed as

$$\lambda^{n+1} = \max \left[0, \frac{A_{PE} - F(\eta^n) + \iint S(\eta^n) \eta^n dx dz}{\iint S(\eta^n) \eta^n dx dz} \right], \quad (B4)$$

where $F(\eta) = \iint f(z, \eta) dx dz$ and $f(z, \eta) = \int_0^\eta [\bar{p}(z - \eta) - \bar{p}(z - \xi)] d\xi$. The new estimate of isopycnal amplitude is computed as

$$\eta^{n+1} = \frac{\lambda^{n+1}}{\lambda^n} \eta^n, \quad (B5)$$

and the new estimate of wave speed is computed as

$$c_0^{n+1} = \sqrt{\frac{gH}{\lambda^{n+1}}}, \quad (B6)$$

where g is the gravitational acceleration. The above procedure is repeated until corresponding convergence criteria are reached. The iteration procedure is transformed to a MATLAB package called Dubreil–Jacotin–Long Equation Solver (DJLES), which is elaborated in Dunphy et al. (2011). The DJLES (code is available at <https://www.math.uwaterloo.ca/~mdunphy/>) is used to find the solution to the DJL equation with the assumption of $U_b = 0$ in this study. The DJLES also needs an input of potential energy which is proportional to the amplitude of mode-1 ISW η . Since η was not observed, the potential energy is determined by fitting the solved vertical velocity w with the w_{isw} obtained from the glider observations. The space-varying solutions η , u , and w in the (x, z) transect can be converted to time- and space-varying solutions in (t, z) by dividing x with the wave speed of ISW.

REFERENCES

- Alford, M. H., R. C. Lien, H. Simmons, J. Klymak, S. Ramp, Y. J. Yang, D. Tang, and M. H. Chang, 2010: Speed and evolution of nonlinear internal waves transiting the South China Sea. *J. Phys. Oceanogr.*, **40**, 1338–1355, <https://doi.org/10.1175/2010JPO4388.1>.
- , and Coauthors, 2015: The formation and fate of internal waves in the South China Sea. *Nature*, **521**, 65–69, <https://doi.org/10.1038/nature14399>.
- Apel, J. R., and Coauthors, 1997: An overview of the 1995 SWARM shallow-water internal wave acoustic scattering experiment. *IEEE J. Oceanic Eng.*, **22**, 465–500, <https://doi.org/10.1109/48.611138>.
- Beaird, N., I. Fer, P. Rhines, and C. Eriksen, 2012: Dissipation of turbulent kinetic energy inferred from Seagliders: An application to the eastern Nordic Seas overflows. *J. Phys. Oceanogr.*, **42**, 2268–2282, <https://doi.org/10.1175/JPO-D-12-094.1>.
- Buijsman, M. C., Y. Kanarska, and J. C. McWilliams, 2010: On the generation and evolution of nonlinear internal waves in the South China Sea. *J. Geophys. Res.*, **115**, C02012, <https://doi.org/10.1029/2009JC005275>.
- Cai, S., X. Long, and Z. Gan, 2002: A numerical study of the generation and propagation of internal solitary waves in the Luzon Strait. *Oceanol. Acta*, **25**, 51–60, [https://doi.org/10.1016/S0399-1784\(02\)01181-7](https://doi.org/10.1016/S0399-1784(02)01181-7).
- Chang, M.-H., R.-C. Lien, Y. J. Yang, T. Y. Tang, and J. Wang, 2008: A composite view of surface signatures and interior properties of nonlinear internal waves: Observations and applications. *J. Atmos. Oceanic Technol.*, **25**, 1218–1227, <https://doi.org/10.1175/2007JTECHO574.1>.
- , —, —, and —, 2011: Nonlinear internal wave properties estimated with moored ADCP measurements. *J. Atmos. Oceanic Technol.*, **28**, 802–815, <https://doi.org/10.1175/2010JTECHO814.1>.
- , and Coauthors, 2021: Direct measurements reveal instabilities and turbulence within large amplitude internal solitary waves beneath the ocean. *Commun. Earth Environ.*, **2**, 15, <https://doi.org/10.1038/s43247-020-00083-6>.
- Chao, S.-Y., D.-S. Ko, R.-C. Lien, and P.-T. Shaw, 2007: Assessing the west ridge of Luzon Strait as an internal wave mediator. *J. Oceanogr.*, **63**, 897–911, <https://doi.org/10.1007/s10872-007-0076-8>.
- Chen, L., Q. Zheng, X. Xiong, Y. Yuan, H. Xie, Y. Guo, L. Yu, and S. Yun, 2019: Dynamic and statistical features of internal solitary waves on the continental slope in the northern South China Sea derived from mooring observations. *J. Geophys. Res. Oceans*, **124**, 4078–4097, <https://doi.org/10.1029/2018JC014843>.
- Dunphy, M., C. Subich, and M. Stastna, 2011: Spectral methods for internal waves: Indistinguishable density profiles and double-humped solitary waves. *Nonlinear Processes Geophys.*, **18**, 351–358, <https://doi.org/10.5194/npg-18-351-2011>.
- Eriksen, C. C., T. J. Osse, R. D. Light, T. Wen, T. W. Lehman, P. L. Sabin, J. W. Ballard, and A. M. Chiodi, 2001: Seaglider: A long-range autonomous underwater vehicle for oceanographic research. *IEEE J. Oceanic Eng.*, **26**, 424–436, <https://doi.org/10.1109/48.972073>.
- Farmer, D. M., M. H. Alford, R.-C. Lien, Y. J. Yang, M.-H. Chang, and Q. Li, 2011: From Luzon Strait to Dongsha Plateau: Stages in the life of an internal wave. *Oceanography*, **24** (4), 64–77, <https://doi.org/10.5670/oceanog.2011.95>.
- Foreman, M. G. G., R. F. Henry, R. A. Walters, and V. A. Balantyne, 1993: A finite element model for tides and resonance along the north coast of British Columbia. *J. Geophys. Res.*, **98**, 2509–2531, <https://doi.org/10.1029/92JC02470>.
- Frajka-Williams, E., C. C. Eriksen, P. B. Rhines, and R. R. Harcourt, 2011: Determining vertical water velocities from Seaglider. *J. Atmos. Oceanic Technol.*, **28**, 1641–1656, <https://doi.org/10.1175/2011JTECHO830.1>.
- Gerkema, T., C. Staquet, and P. Bouuruet-Aubertot, 2006: Decay of semi-diurnal internal-tide beams due to subharmonic resonance. *Geophys. Res. Lett.*, **33**, L08604, <https://doi.org/10.1029/2005GL025105>.
- Grimshaw, R., and K. Helfrich, 2008: Long-time solutions of the Ostrovsky equation. *Stud. Appl. Math.*, **121**, 71–88, <https://doi.org/10.1111/j.1467-9590.2008.00412.x>.
- , and —, 2018: Internal solitary wave generation by tidal flow over topography. *J. Fluid Mech.*, **839**, 387–407, <https://doi.org/10.1017/jfm.2018.21>.
- Helfrich, K. R., and R. H. J. Grimshaw, 2008: Nonlinear disintegration of the internal tide. *J. Phys. Oceanogr.*, **38**, 686–701, <https://doi.org/10.1175/2007JPO3826.1>.
- Huang, X., Z. Chen, W. Zhao, Z. Zhang, C. Zhou, Q. Yang, and J. Tian, 2016: An extreme internal solitary wave event

- observed in the northern South China Sea. *Sci. Rep.*, **6**, 30041, <https://doi.org/10.1038/srep30041>.
- Jackson, C. R., 2009: An empirical model for estimating the geographic location of nonlinear internal solitary waves. *J. Atmos. Oceanic Technol.*, **26**, 2243–2255, <https://doi.org/10.1175/2009JTECHO638.1>.
- Jan, S., and C. T. A. Chen, 2009: Potential biogeochemical effects from vigorous internal tides generated in the Luzon Strait: A case study at the southernmost coast of Taiwan. *J. Geophys. Res.*, **114**, C04021, <https://doi.org/10.1029/2008JC004887>.
- , R.-C. Lien, and C.-H. Ting, 2008: Numerical study of baroclinic tides in Luzon Strait. *J. Oceanogr.*, **64**, 789–802, <https://doi.org/10.1007/s10872-008-0066-5>.
- Klymak, J. M., R. Pinkel, C.-T. Liu, A. K. Liu, and L. David, 2006: Prototypical solitons in the South China Sea. *Geophys. Res. Lett.*, **33**, L11607, <https://doi.org/10.1029/2006GL025932>.
- Ko, D. S., P. J. Martin, C. D. Rowley, and R. H. Preller, 2008: A real-time coastal ocean prediction experiment for MREA04. *J. Mar. Syst.*, **69**, 17–28, <https://doi.org/10.1016/j.jmarsys.2007.02.022>.
- Lamb, K. G., 2003: Shoaling solitary internal waves: On a criterion for the formation of waves with trapped cores. *J. Fluid Mech.*, **478**, 81–100, <https://doi.org/10.1017/S0022112002003269>.
- Ledwell, J. R., L. C. St. Laurent, J. B. Girtton, and J. M. Toole, 2011: Diapycnal mixing in the Antarctic circumpolar current. *J. Phys. Oceanogr.*, **41**, 241–246, <https://doi.org/10.1175/2010JPO4557.1>.
- Li, Q., and D. M. Farmer, 2011: The generation and evolution of nonlinear internal waves in the deep basin of the South China Sea. *J. Phys. Oceanogr.*, **41**, 1345–1363, <https://doi.org/10.1175/2011JPO4587.1>.
- , —, T. F. Duda, and S. Ramp, 2009: Acoustical measurement of nonlinear internal waves using the inverted echo sounder. *J. Atmos. Oceanic Technol.*, **26**, 2228–2242, <https://doi.org/10.1175/2009JTECHO652.1>.
- Lien, R.-C., T. Y. Tang, M.-H. Chang, and E. A. D'Asaro, 2005: Energy of nonlinear internal waves in the South China Sea. *Geophys. Res. Lett.*, **32**, L05615, <https://doi.org/10.1029/2004GL022012>.
- , F. Henyey, B. Ma, and Y. J. Yang, 2014: Large-amplitude internal solitary waves observed in the northern South China Sea: Properties and energetics. *J. Phys. Oceanogr.*, **44**, 1095–1115, <https://doi.org/10.1175/JPO-D-13-088.1>.
- Lund, B., H. C. Graber, J. Xue, and R. Romeiser, 2013: Analysis of internal wave signatures in marine radar data. *IEEE Trans. Geosci. Remote Sens.*, **51**, 4840–4852, <https://doi.org/10.1109/TGRS.2012.2230635>.
- Ma, B., R.-C. Lien, and D. S. Ko, 2013: The variability of internal tides in the northern South China Sea. *J. Oceanogr.*, **69**, 619–630, <https://doi.org/10.1007/s10872-013-0198-0>.
- Ma, W., Y. Wang, S. Yang, S. Wang, and Z. Xue, 2018: Observation of internal solitary waves using an underwater glider in the northern South China Sea. *J. Coastal Res.*, **34**, 1188–1195, <https://doi.org/10.2112/JCOASTRES-D-17-00193.1>.
- Osborne, A. R., T. L. Burch, and R. I. Scarlet, 1978: The influence of internal waves on deep-water drilling. *J. Pet. Technol.*, **30**, 1497–1504, <https://doi.org/10.2118/6913-PA>.
- Pedlosky, J., 2003: *Waves in the Ocean and Atmosphere: Introduction to Wave Dynamics*. Springer, 264 pp., <https://doi.org/10.1007/978-3-662-05131-3>.
- Plant, W. J., W. C. Keller, K. Hayes, G. Chatham, and N. Lederer, 2010: Normalized radar cross section of the sea for backscatter: 2. Modulation by internal waves. *J. Geophys. Res.*, **115**, C09033, <https://doi.org/10.1029/2009JC006079>.
- Ramp, R. S., and Coauthors, 2004: Internal solitons in the northeastern South China Sea. Part I: Source and deep water propagation. *IEEE J. Oceanic Eng.*, **29**, 1157–1181, <https://doi.org/10.1109/JOE.2004.840839>.
- Ramp, S. R., Y.-J. Yang, and F. L. Bahr, 2010: Characterizing the nonlinear internal wave climate in the northeastern South China Sea. *Nonlinear Processes Geophys.*, **17**, 481–498, <https://doi.org/10.5194/npg-17-481-2010>.
- , J.-H. Park, Y. J. Yang, F. L. Bahr, and C. Jeon, 2019: Latitudinal structure of solitons in the South China Sea. *J. Phys. Oceanogr.*, **49**, 1747–1767, <https://doi.org/10.1175/JPO-D-18-0071.1>.
- , and Coauthors, 2022: Solitary waves impinging on an isolated tropical reef: Arrival patterns and wave transformation under shoaling. *J. Geophys. Res. Oceans*, **127**, e2021JC017781, <https://doi.org/10.1029/2021JC017781>.
- Reeder, D. B., B. B. Ma, and Y. J. Yang, 2011: Very large subaqueous sand dunes on the upper continental slope of the South China Sea generated by episodic, shoaling deep-water internal solitary waves. *Mar. Geol.*, **279**, 12–18, <https://doi.org/10.1016/j.margeo.2010.10.009>.
- Robert, C. P., and G. Casella, 2004: *Monte Carlo Statistical Methods*. 2nd ed. Springer, 649 pp., <https://doi.org/10.1007/978-1-4757-4145-2>.
- Rudnick, D. L., T. M. S. Johnston, and J. T. Sherman, 2013: High-frequency internal waves near the Luzon Strait observed by underwater gliders. *J. Geophys. Res. Oceans*, **118**, 774–784, <https://doi.org/10.1002/jgrc.20083>.
- Schmid, C., R. L. Molinari, R. Sabina, Y.-H. Daneshzadeh, X. Xia, E. Forteza, and H. Yang, 2007: The real-time data management system for Argo profiling float operations. *J. Atmos. Oceanic Technol.*, **24**, 1608–1628, <https://doi.org/10.1175/JTECH2070.1>.
- Seber, G. A. F., and C. J. Wild, 2003: *Nonlinear Regression*. Wiley-Interscience, 768 pp.
- Simmons, H., M.-H. Chang, Y.-T. Chang, S.-Y. Chao, O. Fringer, C. R. Jackson, and D. S. Ko, 2011: Modeling and prediction of internal waves in the South China Sea. *Oceanography*, **24** (4), 88–99, <https://doi.org/10.5670/oceanog.2011.97>.
- Sinnett, G., S. R. Ramp, Y. J. Yang, M.-H. Chang, S. Jan, and K. A. Davis, 2022: Large amplitude internal wave transformation into shallow water. *J. Phys. Oceanogr.*, **52**, 2539–2554, <https://doi.org/10.1175/JPO-D-21-0273.1>.
- Stastna, M., 2022: *Internal Waves in the Ocean*. Surveys and Tutorials in the Applied Mathematical Sciences, Vol. 9, Springer International Publishing, 119 pp., <https://doi.org/10.1007/978-3-030-99210-1>.
- , and K. G. Lamb, 2002: Large fully nonlinear internal solitary waves: The effect of background current. *Phys. Fluids*, **14**, 2987–2999, <https://doi.org/10.1063/1.1496510>.
- St. Laurent, L., 2008: Turbulent dissipation on the margins of the South China Sea. *Geophys. Res. Lett.*, **35**, L23615, <https://doi.org/10.1029/2008GL035520>.
- Viúdez, A., 2018: Two modes of vertical velocity in subsurface mesoscale eddies. *J. Geophys. Res. Oceans*, **123**, 3705–3722, <https://doi.org/10.1029/2017JC013735>.
- Vlasenko, V., C. Guo, and N. Stashchuk, 2012: On the mechanism of A-type and B-type internal solitary wave generation in the northern South China Sea. *Deep-Sea Res. I*, **69**, 100–112, <https://doi.org/10.1016/j.dsr.2012.07.004>.
- Wang, S., J. Meng, Q. Li, and X. Chen, 2020: Evolution of internal solitary waves on the slope-shelf topography in the

- northern South China Sea. *Ocean Dyn.*, **70**, 729–743, <https://doi.org/10.1007/s10236-020-01357-5>.
- Wang, Y. H., C. F. Dai, and Y. Y. Chen, 2007: Physical and ecological processes of internal waves on an isolated reef ecosystem in the South China Sea. *Geophys. Res. Lett.*, **34**, L18609, <https://doi.org/10.1029/2007GL030658>.
- Warn-Varnas, A., J. Hawkins, K. G. Lamb, S. Piacsek, S. Chin-Bing, D. King, and G. Burgos, 2010: Solitary wave generation dynamics at Luzon Strait. *Ocean Modell.*, **31**, 9–27, <https://doi.org/10.1016/j.ocemod.2009.08.002>.
- Yang, Y.-J., T. Y. Tang, M. H. Chang, A. K. Liu, M.-K. Hsu, and S. R. Ramp, 2004: Solitons northeast of Tung-Sha Island during the ASIEAX pilot studies. *IEEE J. Oceanic Eng.*, **29**, 1182–1199, <https://doi.org/10.1109/JOE.2004.841424>.
- Zhang, Z., O. B. Fringer, and S. R. Ramp, 2011: Three-dimensional, nonhydrostatic numerical simulation of nonlinear internal wave generation and propagation in the South China Sea. *J. Geophys. Res.*, **116**, C05022, <https://doi.org/10.1029/2010JC006424>.